

Cloner 64

User Handbook

A DNA Cloning Analysis App for macOS

Version 1.5 October 2026

Introduction

Cloner 64 is a DNA cloning analysis application for macOS. It replicates the look and ease of use of Serial Cloner, and runs natively on 64-bit Macs. Cloner 64 reads xdna and xprt files (Strider/Serial Cloner format) as well as FASTA, GenBank, APE, and SnapGene files.

If like me you depended on (and loved using) Christian Marck's DNA Strider and then Serial Cloner by Franck Perez, and were annoyed when Apple decided to no longer support 32-bit apps, then this app might be for you. There are other excellent DNA analysis apps out there for the Mac; however being a creature of habit, I prefer the Serial Cloner approach, and now that my old 32-bit Mac is falling to bits I decided to try and make a version of Serial Cloner that works on the new generation of Macs.

Cloner 64 is possibly not as elegantly crafted as Serial Cloner and does not cover all of the original features - for example there is no Golden Gate cloning, simply because it is not something that I ever did. But there are a few things in this app that complement the original, such as predictive cloning analysis, shuttle vector route-finding, and context-aware methylation checking. The app was created using Anthropic's Claude and DeepSeek to assist with the coding and trouble-shooting.

This handbook covers the main features in Cloner 64 v1.5, to help you get started. Changes are listed under What's New in 1.5 below.

What's New in 1.5

The Predictive Cloning function is improved and now laid out as three numbered steps — where the insert is going (the vector), what is going in (the insert), and how the two should join (the options). The wording throughout has been rewritten to say what each control does in plain language. Out-of-frame fusion strategies are no longer offered, nor are strategies that would cut the tag out of the vector or give a protein shorter than the insert's own ORF. If no enzyme route keeps the frame, the app says so and points you to the PCR strategies, where the primers can add the one or two bases needed. Each fusion strategy now shows how long the resulting protein would be, worked out from the construct that would actually be built.

Cloning fragments are now better interpreted. A fragment annotated from one restriction site to the other — the normal way to annotate what you move between plasmids — no longer has its reading frame taken from its first base. The app finds the reading frames inside it and selects the gene. Reverse-strand inserts have been corrected. A gene on the reverse strand was being measured in the wrong orientation, which would deliver wrong strategies. Flipping a sequence now gives the same answers as before it was flipped, as it should. **Alternative vectors**; when a fusion has no in-frame enzyme route, the results suggest vectors carrying the same tag in a different reading frame such as pET-28a/b/c, pET-21a/b/c, pGEX-4T-1/2/3, pGEX-6P-1/2/3 and pBAD/His A/B/C.

You choose the tag. The app no longer guesses which vector feature is the fusion tag. The list shows every feature with its position, strand and which end of the protein it would sit on, and waits for you to pick. Choosing a feature or an ORF as your insert settles the reading frame, so the second reading-frame picker has gone. It still appears for whole-sequence and custom-position inserts, where the frame genuinely is an open question. Clearer frame wording. Junctions are now described as “5' junction in frame” rather than “In-frame (5')”, because being in frame is a property of the join between tag and insert, not of either sequence on its own.

Scores now carry a description — Simplest, Simple, Moderate, More involved, Most involved — alongside the number. The score ranks what a strategy costs in time, steps and reagents, not whether it will work (it should!).

Feature lengths are corrected. A feature's length was being counted one base too long in the fusion code, which produced a spurious “length not ×3” warning on perfectly good coding sequences and could shift an Insert–Tag fusion out of frame.

Context Help on the results. Every badge on a strategy row — cloning route, partial digest, compatible ends, directionality, orientation, internal cutters, methylation, bases lost and reading frame — now has a context help entry explaining what it means and what to do about it. Tidier strategy rows; View, Save and Print are folded into one Protocol menu, leaving Primers, Build and Verify as the visible actions.

The Restriction Map is improved. Searching now covers both strands and all six reading frames — enter DNA 5' to 3', or protein N to C and Cloner 64 works out the rest, highlighting each hit on the strand it belongs to, with a line under the toolbar saying where the hits are. Protein searches no longer need spaces typed between the residues, matches spanning a line break are found, and the map header says whether it was made from the built-in enzyme library or from My Enzymes only.

Steadier Menu Bar. The menu bar no longer flickers and re-flows when a window opens or context help is switched on.

The Welcome window now tells you when a newer version is available, with a link to the GitHub releases page.

What's New in 1.4

Changing feature colours. You can now recolour a feature without leaving a map. In the Graphical Map, click a feature to select it, then click the colour well in the information bar at the bottom. In the Sequence Map, click List to show the features, then click a feature's colour well. The map updates as you choose. The new colour is saved with the sequence and appears everywhere that feature is shown. Use Undo in the sequence window to go back. Changing a colour in a map does not change the Feature Library, and re-scanning a sequence against the library resets matched features to the library's colours.

Pale feature colours. Files from other programs may use very pale or white feature colours, which are hard to see on a white map. The Sequence Map now shows these darker so the sequence letters stay readable. The Graphical Map shows white or near-white features as light grey instead of blank white boxes. Only the display changes; the colours stored in your files stay the same.

Saving both primers. In Primer Design, Save Both... saves the forward and reverse primers of the selected pair as two .xdna files in a folder you choose.

Protein FASTA files. File → Export Protein → as FASTA saves proteins with the .faa extension, the standard for protein FASTA. Cloner 64 opens .faa files in a protein window. When you open a .fasta file whose sequence contains letters that cannot occur in DNA (such as E, F, I, L, P or Q), Cloner 64 asks whether to open it as protein or DNA.

Export menus. The Export DNA and Export Protein menus are greyed out when they don't match the sequence or protein window in front.

My Enzymes Only. In both the Graphical Map and the Construct Builder, this option now appears under Particular Sites. It only limits the enzyme picker list to your starred enzymes; it does not change which sites are drawn on the map.

Consistent reverse complements. All parts of the program now reverse-complement sequences the same way, including lowercase letters and IUPAC ambiguity codes.

What's New in 1.3

Sequence alignment has been rebuilt. The two alignment windows are merged into one: Align Two Sequences now handles DNA and protein, and its pickers list both kinds of open sequence. Alignments now handle gaps properly — align a cDNA against its genomic sequence, for example, and the introns show up as clean gap blocks. You can align two sequences along their whole length, or ask for just the best-matching region they share. Protein alignments use the standard BLOSUM62 scoring and mark similar amino acids (such as leucine for isoleucine) as well as identical ones.

Two cosmetic issues in the **Restriction Enzyme List** have also been fixed: exporting enzymes no longer doubles the file extension, and enzymes that cut before the end of their recognition site on one strand (such as BsmI) now display their cut offset correctly.

What's New in 1.2

Version 1.2 is almost entirely about the restriction enzyme database.

Enzymes you add or edit are now saved. Previously the enzyme list existed only while the app was running, so anything you added was lost on quit. Your changes are now written to your Library folder and restored on launch.

Only your changes are stored, not a copy of the whole list. The built-in list is rebuilt fresh each launch and your additions, edits and deletions are applied on top. An enzyme you have personally edited keeps your version; everything else stays current.

The Enzyme File menu in the Restriction Enzyme List window writes your enzymes to a file you can back up or send to a colleague. Importing offers a choice of merging with what you already have or replacing it. Restore Defaults returns the built-in list without disturbing your “My Enzymes” selections.

Sixty enzymes with degenerate recognition sites added, including Accl, AflIII, Aval, BanI, BsaJI, BslI, BstEII, BstXI, DraIII, HincII, NlaIV, NspI, ScrFI, SfiI, Styl, XcmI and XmnI. Recognition sequences are taken from REBASE and the NEB catalogue. Degenerate codes are now complemented correctly in the cut structure display.

Degenerate overhangs are no longer treated as fixed sequences. Accl leaves an overhang written MK, which is AG, AT, CG or CT depending on what was cut. Comparing those letters as though they were bases would report two such ends as compatible when they may not anneal at all, so Predictive Cloning and Compatible Cohesive Ends now leave these enzymes out. Enzymes whose overhangs are unambiguous are unaffected — Apol leaves a plain AATT and still pairs with EcoRI.

Two long-standing errors corrected. NlaIII was labelled as leaving a 5' overhang while its stored cut positions described a 3' overhang. I-SceI had its REBASE cut positions recorded without converting them to the form the app uses, which placed the bottom-strand cut on the wrong side of the top-strand cut.

Adding or editing an enzyme now shows both strands as you type, with the cut drawn in place, and refuses entries that contradict themselves — a blunt enzyme whose two cut positions differ, a 5' overhang whose cut positions describe a 3' overhang, or a name already in use. It also warns when a palindromic site is not cut symmetrically, which is the usual sign of a mistyped cut position.

What's New in 1.1

Gel strength in the Virtual Cutter. An agarose slider from 0.5% to 2.0% repositions the bands to where they would run on a gel of that strength, and the co-migration warning now follows it, so you can find a gel that separates a troublesome pair before pouring one.

Residue readout on the hydropathy plot. Hovering the plot reports the residue number, its one-letter code and the hydropathy value at that point.

Type IIS enzymes are no longer offered in Predictive Cloning. BsaI, BsmBI, BbsI and SapI cut outside their recognition site, so their overhang depends on the target sequence and cannot be matched in advance. They remain available throughout the rest of the app.

Corrupt files are now reported on import. Characters in an XDNA or XPRT file that are not valid bases or amino acids used to be accepted silently; an alert now names them.

Corrections to reported positions and fragment sizes. Site positions in the Site Usage window and the Graphical Map are now numbered from 1, matching the Sequence Editor and the rest of the app — these previously read one lower. Digest predictions no longer show a spurious extra band when two enzymes cut at the same position on a circular molecule, which affected isoschizomer pairs such as MspI with HpaII.

Getting Started

Opening Cloner 64 for the First Time

Cloner 64 is not signed with an Apple Developer certificate, so macOS blocks it the first time you try to open it. This is a one-time step and the app works normally afterwards.

On macOS 15 (Sequoia) and macOS 26 (Tahoe) the app may appear to start — an icon shows in the Dock — but no window ever opens and no warning is displayed. That is macOS silently blocking it, not a fault in the app.

First try the simple route: move Cloner 64 to your Applications folder, right-click it and choose Open, then click Open again if a warning appears. On macOS 14 and earlier that is all that is needed.

If the app still will not open, it's because Apple removed that shortcut in macOS 15 and one Terminal command is required. Open Terminal (Applications → Utilities → Terminal) and type the following, *including the space at the end*, without pressing Return:

```
xattr -dr com.apple.quarantine
```

Then drag Cloner 64 from Finder into the Terminal window, which fills in the location for you, and press Return. If nothing is printed, it worked — open the app normally. Should you see “Operation not permitted”, switch on Terminal under System Settings → Privacy & Security → App Management and repeat.

macOS tags every downloaded file with a quarantine marker. That command removes this marker from Cloner 64 only; it changes no security settings and affects no other application.

Opening Files

Cloner 64 supports the following file formats:

Format / Extensions	Notes
XDNA (.xdna)	Serial Cloner DNA format (read/write)
XPRT (.xprt)	Serial Cloner protein format (read/write)
FASTA (.fasta, .fa, .fna, .faa)	Plain sequence (read/write). (.faa = protein)
GenBank (.gb, .gbk)	Annotated sequence (read/write)
APE (.ape)	A Plasmid Editor format (read/write)
SnapGene (.dna)	SnapGene binary format (read)

To open a file, double click on it, or use File › Open or drag files onto the app icon in the Dock. Recently opened files appear under File › Open Recent. SnapGene files are detected automatically by their binary signature.

Creating a New Sequence

Choose File › New › DNA Sequence to create an empty sequence, or File › New › Protein Sequence for a protein. You can also paste a sequence from the clipboard: Edit › Paste as New Sequence (Cmd+Shift+V).

Saving and Exporting

Save your work with File > Save. Use File > Save As to save a copy with a new name. Export menus let you save DNA sequences as FASTA, GenBank, or APE, and protein sequences as FASTA (.faa) or XPRT.

The Sequence Editor

The Sequence Editor is your main workspace. It displays the DNA sequence with line numbers, complementary strand for DNA, and colour-coded features.

Locked and Unlocked State

Sequences opened from disk are locked by default to prevent accidental edits. A “Locked” checkbox in the editor toolbar shows the current state. Uncheck it to enable editing.

Editing

Click anywhere in the sequence to position the cursor, then type to insert bases. Select a region by clicking and dragging, or use Shift+Click to extend a selection. Use the main Edit and Tools menus for copy, cut, paste, translation, and other options.

Features

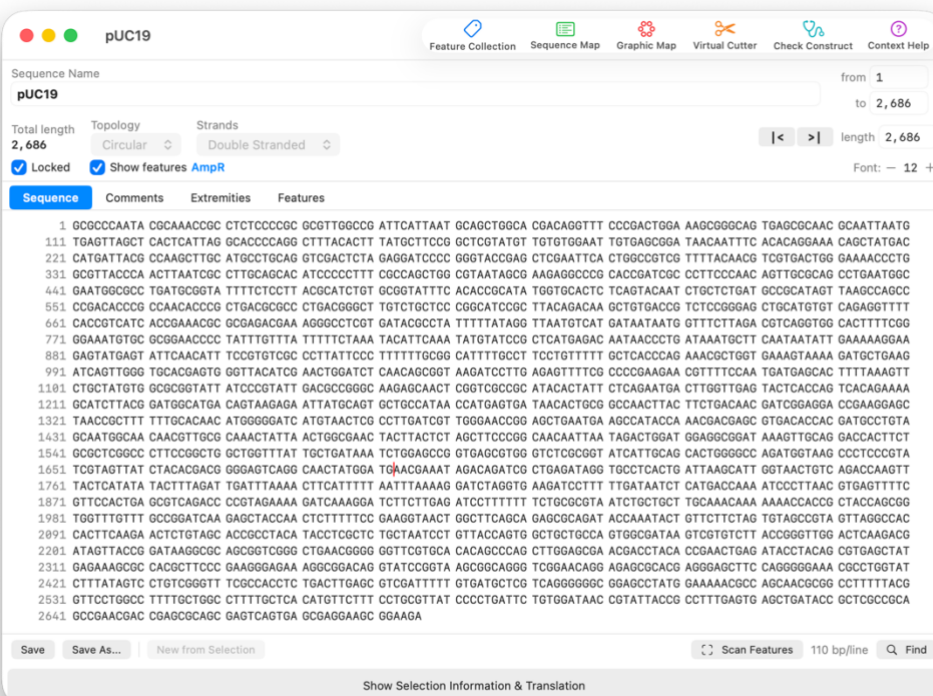
Annotated features (promoters, genes, origins, etc.) appear as coloured bars overlaid on the sequence. These are embedded in the .xdna sequence file or may be added from the Feature Collection by a search. Each feature has a name, type, location, and reading direction. From the Features tab you can edit individual features. Features that are added to the file are preserved upon saving the file.

Translation

Select a region of the sequence and choose Tools > Translate Selection to translate it into a protein sequence. The translated protein opens in its own window. Stop codons are shown as asterisks (*). Or click on “show selection information and translation”

Find Drawer

The Find drawer lets you search for specific sequences, restriction enzyme sites, and open reading frames (ORFs) within your sequence.



Sequence Editor View

Clicking Find searches for sequences, restriction sites, and ORFs and opens on the right.

Translations and general information open underneath the sequence

Graphical Map

The Graphical Map displays your sequence as a circular or linear map, depending on sequence topology.

Restriction Enzyme Sites

Use the Sites toolbar button (Scissors) to choose which restriction sites to display:

- Unique Sites - enzymes that cut exactly once (cream-coloured)
- Double Sites - enzymes that cut exactly twice (brown-coloured)
- Blunt Sites - blunt-end cutters (teal-coloured)
- Particular Sites - choose specific enzymes from a list (blue)
- My Enzymes Only limits the picker list to starred enzymes

Display Options

The Display button (Eye icon) lets you toggle features and ORFs, and selectively show or hide individual features and ORFs. The colours of features can be changed from the info bar (click a feature to show this). The Split icon will give a view of the sequence below the map.

Methylation Sensitivity

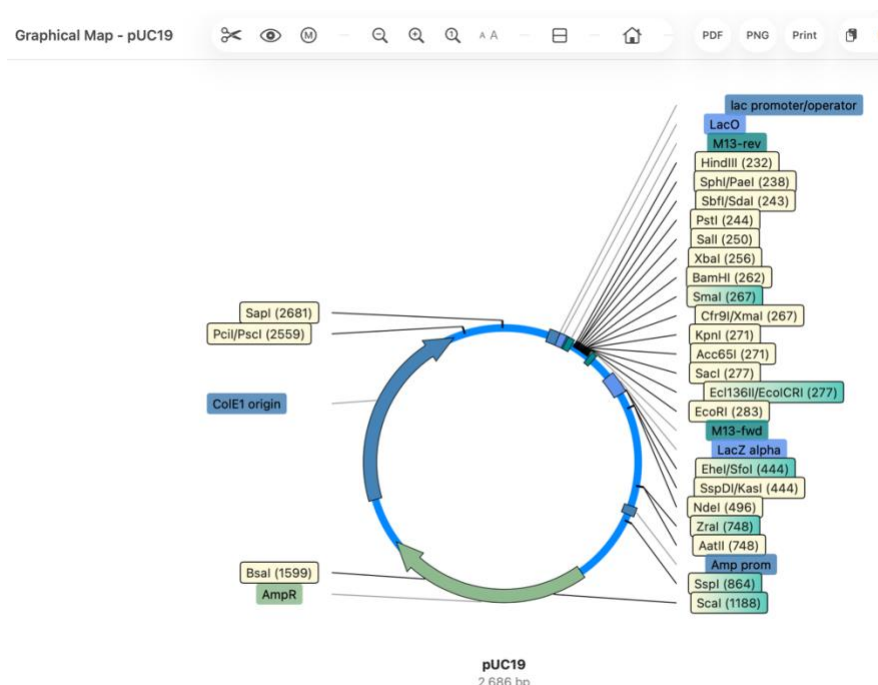
The methylation button (M icon) controls Dam, Dcm, and CpG methylation sensitivity. The app checks the sequence context at each cut site, so methylation warnings are only shown where the relevant methylation motif is present. Affected sites are shown in red on the map.

Zoom and Font Size

Use the magnifying glass buttons to zoom in and out. The text size buttons (A smaller / A larger) adjust label font size. Press the (1) button to reset zoom to 100%.

Export and Print

Export your map as PDF or PNG using the toolbar buttons. Use Print to send the map to a printer.



Graphic Map

Clicking on features allows you to select them.

Clicking on two restriction sites allows you to select the fragment(s)

Sequence Map

The Sequence Map (accessible from the Map icon in the Sequence Editor) displays a text-based restriction map of your sequence. Enzyme cut sites are marked directly on the sequence text, with optional translation frames for both forward and reverse reading frames shown above and below the DNA sequence. Feature annotations are displayed in context, and can be listed and colour edited. You can configure the line width, and export the map via copy, or print. Search accepts either DNA or protein. A DNA search also finds the sequence on the reverse strand. A protein search is matched against the translation, so type the residues straight through — FSTYA, not F S T Y A — and it will be found on forward or reverse reading frames.

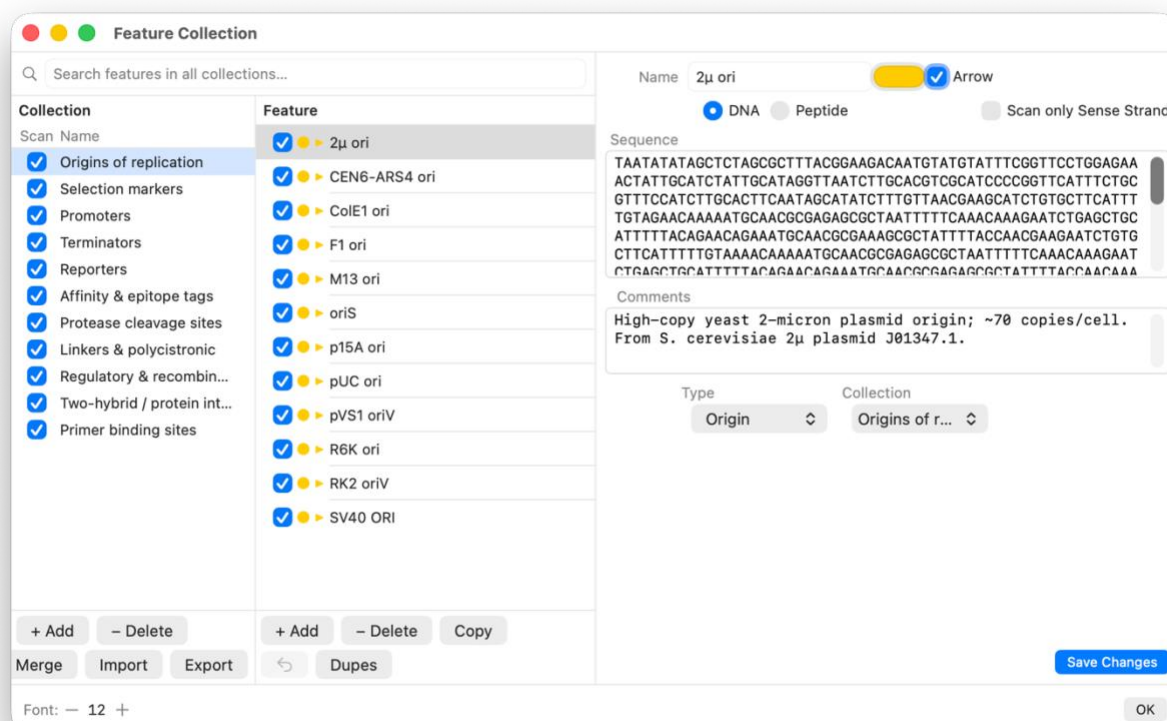
Searching the map: The search box finds a DNA or an amino acid sequence anywhere in the map and highlights it. Type the sequence as DNA 5' to 3', or protein from N-terminus to C-terminus with no spaces between the letters. Both strands and all six reading frames are searched. A match on the bottom DNA strand is highlighted on the bottom-strand row, and a match in a reverse reading frame on that frame's row. Because those rows are read from right to left, a reverse-strand match reads backwards across the page. A line under the toolbar says where the hits are: which strand, and which reading frame. A palindromic restriction site reads the same on both strands, so it is reported once and labelled palindromic rather than counted twice. Overlapping occurrences each count as a separate hit, since each is a real position. Matches are found whether or not they fall across a line break, and whether or not **Show codons** is switched on.

The header records which enzymes the map was made from: the built-in library, or your My Enzymes list.

Feature Collection

The Feature Collection (Tools > Feature Collection, or from the Label icon in the Sequence Editor) lets you view, edit, add, and remove annotated features (DNA or protein) which can be noted on your sequence by scanning. You can reorder features by dragging, and the collection can be screened for duplicates.

Use Tools > Scan Sequence for Features, or the Scan Features button below the sequence to automatically scan your sequence against the built-in Feature Library. The library contains common molecular biology elements organised into eleven collections — including origins of replication, selection markers, promoters, terminators, reporters, epitope tags, protease cleavage sites, and recombination sites. You can also create, import, and export your own feature collections.



Tools

Reverse and Complement

Tools › Reverse Complement reverses the entire sequence and takes its complement ($A \leftrightarrow T$, $G \leftrightarrow C$). This is useful when you need to work with the opposite strand. Separate Reverse and Complement commands are also available.

RNA/DNA Conversion

Convert between DNA and RNA using Tools › Convert to RNA ($T \rightarrow U$) and Convert to DNA ($U \rightarrow T$).

Site Usage

Tools › Site Usage opens a table showing all restriction enzyme recognition sites found in your sequence, including cut positions, fragment sizes, and whether each site is affected by methylation. You can filter by unique cutters, blunt cutters or absent cutters as well as restricting the list to your own enzyme stock.

Restriction Enzyme List

Tools › Restriction Enzyme List opens a searchable reference of all restriction enzymes in the database, showing recognition sequences, cut positions, overhang types, and methylation sensitivity. Isoschizomers with same recognition site and cut position are consolidated into combined entries (e.g. *AflIII*/*BspTI*). You can add, edit, or remove enzymes, and your changes are saved automatically.

Cut positions are given as offsets from the start of the recognition site. For a Type IIS enzyme, which cuts outside its site, they are shown in the catalogue form instead — *BsaI* reads +1 and +5, which matches the catalogue format.

The cut structure column draws the recognition site split at the cleavage point on both strands, so the length and polarity of the overhang can be read directly. For an enzyme with ambiguity codes in its site, the lower strand shows the complementary codes: *AccI* (GTMKAC) reads CAKMTG, because M pairs with K.

Enzyme File → Export My Enzymes writes the enzymes you have added, edited or deleted to a file. This is worth doing once you have built up a list: it can be shared with another Cloner 64 user, or kept as a backup. Import offers to merge the file with your current list or replace it. Restore Built-in Defaults discards your changes and returns the supplied list.

Your enzymes survive installing a new version of Cloner 64. They are kept separately from the app itself, so replacing the application does not disturb them. They would only be lost if you deleted the Cloner 64 folder inside your Library's Application Support folder, or moved to a machine that had never run the app — which is the case the export file covers.

My Enzymes

Within the Restriction Enzyme List you can earmark enzymes that you actually have in stock. Click the star (★) beside any enzyme to add it to your “My Enzymes” collection. Use the My Enzymes toggle in the toolbar to filter the list to only your starred enzymes. Your selection is saved between sessions and survives installing a new version. The My Enzymes filter carries through to other parts of the app: for example Check Construct can confine its digest suggestions to your starred enzymes only, saving you from being offered enzymes you don't have.

Compatible Cohesive Ends

Tools › Compatible Cohesive Ends shows which enzyme pairs produce compatible sticky ends (e.g. BamHI and BglII), which is essential for planning cloning strategies. The table is computed dynamically from the current enzyme database, and can be searched by enzyme name or overhang site.

Two classes of enzyme are deliberately absent. Type IIS enzymes cut outside their recognition site, and enzymes with ambiguity codes in their site leave an overhang whose bases depend on the sequence cut. In both cases the overhang is a property of the target rather than of the enzyme, so there is no answer to give in the abstract. To check a specific case, cut the actual sequence and compare the overhangs reported in Site Usage.

Cloning Vector Library

Tools › Cloning Vector Library opens the Shuttle Vector Library, which contains a metadata database (not the full sequence) of common vectors with their MCS enzyme information, selection markers, sizes, and categories. You can add your own vectors and import/export entries. Each vector in the library can be earmarked as one of yours using the star (★) button. Use the My Vectors toggle in the toolbar to filter the list to only your starred vectors. Your selection is saved between sessions and is used by the Predictive Cloning shuttle route search (see below).

Where a vector belongs to a family that covers all three reading frames — pET-28a/b/c, pET-21a/b/c, pGEX-4T-1/2/3, pGEX-6P-1/2/3 and pBAD/His A/B/C — each member is listed with the frame it uses. Predictive Cloning draws on this when a fusion has no in-frame route, to suggest a sibling vector that may fit.

Reference Tables

Quick reference tables for the standard genetic code and IUPAC nucleotide ambiguity codes are available under Tools › Genetic Code and Tools › IUPAC Nucleotide Codes.

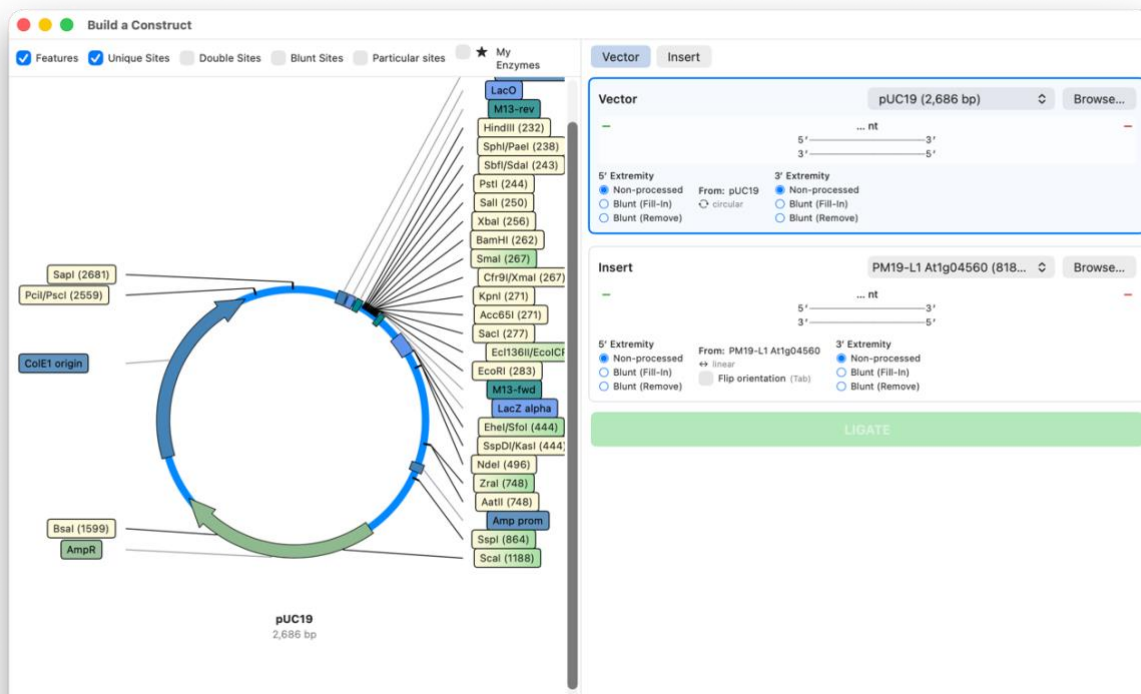
Functions

Build a Construct

Function › Build a Construct opens the Construct Builder for in silico ligation. Select a vector and insert, choose restriction enzyme cut sites on the graphical maps, and Cloner 64 will simulate the ligation, checking sticky-end compatibility and fragment orientation. The resulting construct can be saved as a new sequence.

Key features of the Construct Builder:

- Vector and Insert panels showing maps and details of 3' and 5' ends
- Sticky-end compatibility matching across enzymes
- Backbone and insert fragment choice using enzyme cut positions
- My Enzymes Only limits the picker list to starred enzymes
- Overhang display allows end processing options (fill/trim)
- Handles both circular and linear sequences



Build a construct. Select restriction sites in the vector and in the insert, and press ligate. You can save the new construct or carry out a fresh ligation with different restriction sites.

Verify Construct

The Verify Construct function is made available by the Predictive Cloning and Construct Builder tools after a construct has been assembled. It provides a verification digest report for that specific construct, taking into account the insert position and length. Two modes are available: **Insert orientation** must be determined - tick this when the ligation was non-directional and you need a digest that distinguishes forward from reverse insert. The tool looks for enzymes that cut the insert asymmetrically so the two orientations give different band patterns. **Orientation fixed** - untick this when the cloning strategy already forces the orientation (for example, a directional two-enzyme ligation). The tool instead looks for flanking diagnostic digests that confirm insert presence.

Check Construct

Function › Check Construct or the Stethoscope icon in the Sequence Editor window opens the diagnostic digest advisor. Given a construct (or any sequence), it suggests restriction digest strategies to confirm that your plasmid is what you think it is or that your cloning worked. There are four modes:

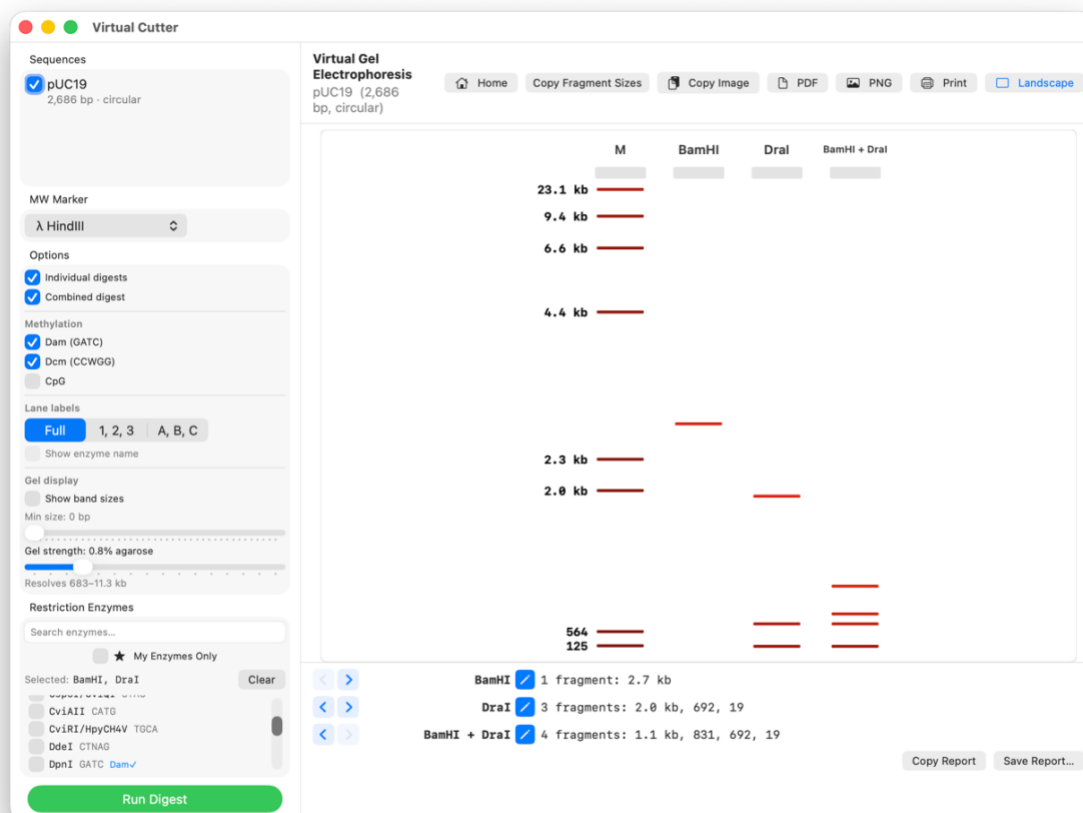
- Fingerprint mode - finds enzymes that give a distinctive gel pattern for the whole construct
- Presence mode - finds enzymes confirming that a feature is present
- Orientation mode - finds enzymes that can distinguish forward from reverse insert orientation
- Comparison mode to distinguish recombinants from parent vector

Virtual Cutter

Function › Virtual Cutter, or Scissors icon from the Sequence Editor, performs a virtual restriction digest. Choose one or more enzymes and the app simulates the digest, showing fragment sizes and a gel electrophoresis visualisation. Results can be exported as PDF, PNG, or printed.

The gel strength slider sets the agarose concentration, from 0.5% to 2.0%, and moves the bands to where they would run on a gel of that strength. Lower percentages spread out large fragments, higher percentages spread out small ones. Fragments above the range bunch up near the well, and those below it run down near the dye front, so the same digest can look quite different on two gels. The caption under the slider shows the range the current setting resolves.

The co-migration warning on each lane follows the slider. Two fragments that merge into a single band on a 2% gel may separate cleanly on a 0.7% one, and if a different strength would resolve more bands a note appears suggesting it. Exact spacing may depend on your buffer, voltage and run time.



Predictive Cloning

Function › Predictive Cloning opens the automated cloning strategy analyser. You give it a vector and an insert; it tries every enzyme combination it can and ranks the ones that work. The window is laid out as three steps.

Step 1 — Where is it going? (vector)

Choose the vector, and say where in it the insert should land. Insert into offers By position, where you give a start and end — typically the MCS — or Between two features, where you name the features either side. Leave the positions blank to search the whole plasmid. Detect MCS will make a sensible guess at the polylinker for you.

If the vector is in the Cloning Vector Library its cloning sites are already known and shuttle routes can be searched. If not, every enzyme site is checked instead.

Protected regions lists features and ORFs in the vector that you may not want disturbed; anything ticked is kept intact and strategies that would cut it are dropped. Regions that overlap the site you are cloning into are shown greyed out and are never protected — otherwise you would be protecting the very place you are trying to cut.

Step 2 — What are you putting in? (insert)

Choose the source sequence and then what part of it to use: the whole sequence, an annotated feature, an ORF, or custom positions. Find a feature by name searches every open sequence for a feature of that name and analyses each match in turn — useful when you have several candidate constructs for the same gene.

In a fusion, choosing a feature or an ORF also settles the reading frame and the strand, and a line underneath confirms it. If the feature is on the reverse strand the insert is reverse-complemented before fusion, and the line says so. Custom positions is the route to take when you want only part of an ORF, or a stretch spanning more than one.

Fragments are normally annotated from one restriction site to the other, because that is what you cut out and move between plasmids. The coding sequence therefore starts somewhere inside the fragment, not at its first base. When that is the case the app says so, finds the reading frames within the fragment and selects the longest — which is almost always the gene being moved. The other frames stay available under Step 3 in case the one you want is different; the amino-acid count is the best guide.

Step 3 — How should it join? (options)

Simple insertion puts the insert into the vector and does not check the reading frame. The fusion modes join the insert to a tag already in the vector — a His-tag, GFP, GST and so on — to make one protein. The names read from the start of the protein to the end: Tag–Insert puts the tag at the N-terminus, Insert–Tag at the C-terminus, Tag–Insert–Tag at both.

For a fusion you choose which vector feature is the tag. The app does not guess: the list shows every feature with its coordinates, strand and, where it can work it out, which end of the protein that tag would sit on. Once a tag is chosen the reading frame at each junction is worked out for you, and the frame offset boxes stay folded away under Adjust frame offsets manually. You only need them if the tag is not annotated as a feature.

Use only My Enzymes restricts the search to the enzymes you have starred as freezer stock.

Reading the results

Strategies are grouped into Direct Digest, PCR and, where the insert is already a blunt fragment, Blunt Insert. Each row is scored, and each carries badges summarising what the strategy involves. Hover over any badge with context help switched on for an explanation. The badges cover:

- Cloning route — direct digest, blunt via fill-in or nibble-back, or PCR.
- Directionality — whether the insert can only go in one way round, or either way.
- Reading frame — in fusions, whether each junction keeps the tag and insert in the same frame.
- Internal cutters — whether an enzyme also cuts inside the insert.
- Methylation — whether Dam, Dcm or CpG methylation blocks a site, or is required by it.
- Bases lost — where the cut does not sit exactly at the insert boundary.

The score ranks what a strategy costs you — time, hands-on steps, reagents and the number of things that can go wrong — not whether it will work. Every strategy listed should give you the construct. The number is accompanied by a description, from *Simplest* through to *Most involved*.

In a fusion, each row also shows the predicted length of the fusion protein — counted from the start codon that begins the tagged protein through to the first stop codon, using the construct the Build button would actually make. It is usually the most informative number on the row: one much shorter than you expect means the insert is being read in the wrong frame, or a stop codon lies between the tag and your coding sequence.

Each row offers Primers (for PCR routes), Build, Verify and a Protocol menu holding View, Save and Print. Build generates the predicted product as a new sequence with its features remapped; Verify suggests a diagnostic digest for screening colonies.

Fusion cloning and the reading frame

In a fusion the tag and the insert must be read in the same frame, so the protein runs from one into the other without a frameshift. Where the enzyme cuts decides whether the two line up. The app checks this at every junction for every candidate enzyme.

A junction that is out of frame does not give a poorer fusion protein — it gives none at all. Such strategies are therefore left out of the results rather than ranked low. If no enzyme route keeps the frame, the app says so plainly and points you to the PCR strategies, where the primers can add the one or two extra bases needed to set it.

Three further checks remove strategies that could not give the construct: those whose two vector cuts would excise the tag itself, leaving nothing to fuse to; those whose predicted protein is shorter than the ORF being fused, meaning translation stopped before or inside it; and those whose protein cannot be worked out at all. The principle throughout is that a strategy offered in a fusion should produce a fusion protein.

It may also suggest alternative vectors. Many expression vectors come as a set of two or three that are identical apart from the spacing between the tag and the cloning site, so between them they cover all three reading frames for example pET-28a/b/c, pET-21a/b/c, pGEX-4T-1/2/3, pGEX-6P-1/2/3 and pBAD/His A/B/C. If your insert will not go into one of them in frame, one of its siblings usually will. The Cloning Vector Library holds each vector's details but not its sequence, so the app can point you at the siblings but cannot prove any

one of them will work. Obtain that vector's sequence, open it in Cloner 64, select it and run the analysis again.

Insert direction preference

Use the Insert Direction control to specify whether the insert should go in forward, reverse, or either orientation. This narrows the results when orientation matters.

Shuttle vector routes

If no strategy works with your chosen vector, the Shuttle Routes tab searches for PCR-free multi-step routes through intermediate vectors, finding a chain of compatible plasmids to move the insert along step by step. The Cloning Vector Library holds metadata only — MCS enzymes, markers, sizes and categories — not full sequences, so once a route looks promising it should be checked against the real shuttle vector sequence.

Type IIS enzymes

Enzymes that cut outside their recognition site — BsaI, BsmBI, BbsI and SapI — are not offered here. Every strategy this tool builds depends on knowing an enzyme's overhang in advance so it can judge whether two cut ends will join, and a Type IIS overhang is whatever sequence happens to lie at the cut, a property of the target rather than of the enzyme. Golden Gate assembly, where that is the whole point, would need a different approach and is not currently supported. These enzymes remain available everywhere else in the app, including the Virtual Cutter and the site maps.

Design PCR Primers

Function › Design PCR Primers opens the primer design tool. Select a template sequence, define your target region (or pick a feature/ORF), and the tool suggests forward and reverse primers with calculated melting temperatures and GC content. Primers can include 5' tails for adding restriction sites. The visual template map has draggable handles for adjusting the target region. Designed primers (single or pairs) can be saved and imported with tail/annealing annotations preserved. Primer-dimer screening is built in. Circular templates are supported.

Primer Stock

If you have a folder of existing primer files (.xdna format), you can nominate it as your Primer Stock folder in the Primer Design tool. The app will automatically screen your stock primers against the current template and, where a match is found, use the stock primer as a weighted candidate in the design results. This saves re-ordering primers you already have.

Run a PCR

Function › Run a PCR simulates a PCR reaction. Choose a template, provide forward and reverse primers, select a polymerase (Taq or Pfu/Phusion), and the tool predicts the amplified product including any 5' tails and Taq A-overhangs. The product can be saved as a new sequence.

Sequence Alignment

Function › Align Two Sequences performs pairwise alignment of any two open sequences, DNA or protein. The pickers list DNA sequences first, then protein sequences (marked “(protein)”), with the length — and for proteins the molecular weight — shown beneath.

The Mode control sets the scoring: Auto detects the sequence type, DNA uses match/mismatch scoring, and Protein uses the BLOSUM62 substitution matrix. The Scope control chooses between Full length, which keeps both sequences end to end so that long insertions such as introns appear as continuous gap blocks (for example when aligning genomic DNA against a cDNA), and Local, which trims the result to the best-matching shared region (Smith–Waterman), useful for finding a shared domain between otherwise different sequences. Position numbers always refer to the original sequences, so a local alignment may start partway in.

For DNA the match line marks identical positions with |. For proteins it shows * (identical), : (conservative) and . (semi-conservative), identity and similarity percentages are both reported, and the Color coded option colours residues by their chemical class.

NCBI BLAST Search

Function › NCBI BLAST Search DNA and BLAST Search Protein open the NCBI BLAST web interface pre-loaded with your current sequence for online homology searching.

Hydropathy Plot

Function › Hydropathy Plot generates a Kyte-Doolittle hydropathy plot for the current protein sequence, useful for predicting transmembrane regions. The transmembrane threshold (set at 1.6, but modifiable) is shown as a reference line. The plot is zoomable.

Moving the pointer across the plot shows a crosshair with the residue number, its one-letter code and the smoothed hydropathy value at that point. Note that the curve begins and ends short of the sequence itself: each point is the average of a window of residues centred on it, so with the default window of 9 the plot runs from residue 5 to the fifth residue from the end. Widening the window increases that margin, and the plot is directly comparable with ProtScale or EMBOSS output for the same protein and window.

Protein Sequences

Cloner 64 supports protein sequences with a dedicated viewer. Protein windows show the amino acid sequence with Clustal-style colour-coded residues and a legend. A properties panel shows molecular weight, isoelectric point (pI), and extinction coefficient. Proteins can be opened from XPRT files, protein FASTA (.faa) files or created by translating a DNA selection.

Keyboard Shortcuts

Shortcut	Action	Shortcut	Action
Cmd+N	New DNA Sequence	Cmd+Shift+N	New Protein Sequence
Cmd+O	Open File	Cmd+S	Save
Cmd+Shift+S	Save As	Cmd+Z	Undo
Cmd+Shift+Z	Redo	Cmd+X/C/V	Cut / Copy / Paste
Cmd+Shift+V	Paste as New Sequence	Cmd+A	Select All
Cmd+U	Make Uppercase	Cmd+Shift+U	Make Lowercase
Cmd+T	Translate Selection	Cmd+L	Feature Collection
Cmd+B	Scan for Features	Cmd+Shift+K	Build a Construct
Cmd+Shift+D	Virtual Cutter	Cmd+Shift+R	Run a PCR
Cmd+Shift+A	Align Two Sequences	Cmd+Shift+E	Export DNA as FASTA
Cmd+P	Print	Cmd+Shift+P	Page Setup
Cmd+Shift+?	Context Help on/off	Escape	Exit Full Screen

Tips and Tricks

- Use the Find drawer to locate restriction sites, ORFs, and specific sequence motifs.
- Double-click a feature on the graphical map to open it as a new sequence window; single-click selects it and shows the info bar
- When designing cloning strategies, check Tools › Compatible Cohesive Ends to find enzyme pairs that produce compatible sticky ends.
- Use Predictive Cloning to automatically screen all possible enzyme combinations for a vector/insert pair - it saves a lot of manual checking.
- The Primer Design tool can target a specific feature or ORF - use the target picker dropdown.
- PCR products from the PCR Simulation can be used directly as inserts in the Construct Builder.
- Methylation sensitivity is context-aware; the app checks the actual sequence context at each cut site rather than just flagging all sites for a sensitive enzyme.
- Star (★) your commonly used enzymes in the Restriction Enzyme List to build a My Enzymes set. Use the My Enzymes filter in Check Construct to get suggestions restricted to your own freezer stock.
- Star (★) the vectors you have in your lab in the Cloning Vector Library. Enable My Vectors Only in the Shuttle Routes tab to restrict the pathfinder to routes you can actually carry out.
- If you have a folder of existing primer files, nominate it as your Primer Stock in Primer Design. The app will automatically find and prefer matching primers you already have.
- The Shuttle Vector Library pathfinder can identify multi-step PCR-free routes between vectors that share compatible MCS sites.
- Context Help is available throughout the app. Turn on Context help from the main Help menu, then hover over any button, toggle, picker, or field for an explanation of what it does and when to use it.

Cloner 64 -Built for molecular biologists who miss Serial Cloner on modern Macs.